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Seismic Image Denoising with Zero Shot Strategy

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Abstract

We propose a data-free seismic denoising network based on the ideas of Zero-Shot Noise2Noise from pioneering deep learning techniques for natural image processing. The idea involves generating a pair of noisy images from a single noisy image and subsequently training a small network on this pair. The method engages the use of a downsampling operator to decompose an image into a pair of downsampled images. A lightweight network is subsequently trained using regularization techniques to map one downsampled image to another. The loss function consists of two terms: residual loss and consistency loss. The residual loss in denoising neural networks is formulated using only the downsampled images as input, while the network receives the image in full spatial resolution for the consistency loss. The key insight of this network is that the network can learn to denoise seismic images without any training data or knowledge of the noise distribution. This is achieved by training the network in a self-supervised manner. Numerical experiments demonstrate the validity of the network on seismic data.

Introduction

Background and Motivation

Seismic imaging is a cornerstone of hydrocarbon exploration, providing subsurface structural and stratigraphic information. However, seismic data acquired in field surveys are invariably contaminated by noise arising from environmental factors, instrumentation errors, and processing artifacts. Random noise obscures subtle geological features, complicating interpretation and reducing the reliability of reservoir characterization. Effective denoising is thus essential to enhance data quality while preserving critical signals such as faults, reflectors, and amplitude variations.

Challenges in Traditional Methods

Seismic image denoising is a process that involves the suppression of random noise in seismic data while preserving geological signals. Various methods have been developed over the years to attenuate noise, including frequency-domain methods (e.g., f-x deconvolution), transform-domain methods (e.g., wavelet and curvelet transform), and sparse representation methods. Frequency-domain methods exploit signal predictability in the frequency-space domain. Transform-domain methods separate noise and signal in

multi-scale representations. Sparse representation methods leverage compressed sensing principles. While these techniques have been widely adopted, they are not without challenges. For example, frequency-domain methods are still sensitive to parameter selection (e.g., filter lengths, thresholding levels), while the effectiveness of transform-based denoising methods depends on the choice of base function. In addition, maintaining the fidelity of seismic data by preserving signals that reflect geological information is a major challenge. As such, developing effective methods to suppress random noise while minimizing signal loss is critical for improving the signal-to-noise ratio (SNR).

Deep Learning in Seismic Denoising

Deep learning has shown remarkable efficacy in suppressing random noise due to its ability to learn priors end-to-end, bypassing complex manual design. This is because deep learning enables the direct acquisition of prior knowledge through end-to-end learning, which eliminates the need for complex prior design. Furthermore, deep learning frameworks are amenable to parallel computing, thereby facilitating their deployment in seismic denoising applications. So deep learning is a widely utilized approach that allows for the efficient and effective suppression of random noise.

The conventional deep learning method for noise reduction employs supervised learning, training a neural network with noisy data as input and clean data as labels. However, this approach has limitations when applied to field data, as it is difficult to obtain clean seismic data from observed seismic data. For this reason, supervised learning often uses the traditional denoising methods on real data or tries adding noises to synthetic data, to construct training datasets for field data (Zhang et al., 2017; Yu et al., 2019). These approaches introduce biases and limit generalization to real-world noise.

Unsupervised deep learning techniques have also been successfully developed in the attenuation of noise in seismic datasets. This approach has yielded promising results in reducing unwanted noise and enhancing the accuracy of data analysis and interpretation. The application of unsupervised deep learning algorithms to seismic denoising has demonstrated its potential as a valuable tool for seismic data processing (Lehtinen et al., 2018; Liu et al., 2021; Wang et al., 2023).

Contributions

In this paper, we proposed a data-free seismic denoising network based on the ideas of Zero-Shot Noise2Noise from pioneering deep learning techniques for natural image processing (Mansour and Heckel, 2023). The key insight of this network is that the network can learn to denoise seismic images without any training data or knowledge of the noise distribution. This is achieved by training the network in a self-supervised manner. Numerical experiments demonstrate the validity of the network on seismic data.

This study makes the following contributions:

1. Adaptation of Zero-Shot N2N to seismic data: Customization of the framework for seismic image characteristics, including non-stationary noise and curved events.
2. Lightweight network design: A shallow CNN with ~20k parameters, optimized for computational efficiency and compatibility with field data constraints.
3. Comprehensive validation: Quantitative and qualitative comparisons against FXDecon on synthetic datasets, emphasizing SNR improvements and structural preservation.

Methodology

Problem Formulation

Seismic denoising aims to recover a clean image x from a noisy observation $y = x + n$, where n represents additive random noise. The objective is to learn a denoising operator f_θ parameterized by θ , such that $\hat{x} = f_\theta(y)$ minimizes the mean squared error (MSE) $E[\|x - \hat{x}\|^2]$.

From Noise2Noise to Zero-Shot Learning

Generally, the deep learning-based denoising methods are typically neural networks that map a noisy data to an estimation of the clean data. Neural networks can also be trained on different noisy observations of the same clean data. However, Zero-Shot Noise2Noise is a deep learning technique that has been developed based on two other techniques, namely Noise2Noise and neighbor2neighbor. The Noise2Noise technique involves training a neural network using a pair of noisy images, while the neighbor2neighbor technique uses a single noisy image to generate a pair of noisy maps. Zero-Shot Noise2Noise builds upon these techniques to further enhance the quality of denoising. The idea involves generating a pair of noisy images from a single noisy image and subsequently training a small network on this pair. Specifically, the method engages the use of a downsampling operator to decompose an image into a pair of downsampled images. A lightweight network is subsequently trained using regularization techniques to map one downsampled image to another. This training process is designed to enable the network to learn the underlying structure and features of the image (Mansour and Heckel, 2023).

Loss Function and Consistency Regularization

To mitigate overfitting during the training of Zero-Shot denoising network from one image, a shallower architecture is employed along with an explicit regularizer. The regularizer is specifically designed to encourage the model to explore different characteristics. By incorporating these measures, the proposed methodology seeks to address the issue of overfitting that can arise during training. The use of a shallower architecture may also result in faster training times and improved performance. Overall, this approach presents a promising solution for training Zero-Shot network from one image that warrants further investigation.

The loss function consists of two terms: residual loss and consistency loss. The residual loss in denoising neural networks is formulated using only the downsampled images as input, while the network receives the image in full spatial resolution for the consistency loss. Incorporating the consistency loss yields a twofold benefit of enhancing the denoising performance and preventing overfitting and thus serves as a regularizing term. Let D_1 and D_2 denote non-overlapping downsampling operators. The total loss is:

$$L(\theta) = L_{res}(\theta) + L_{cons}(\theta) \quad (1)$$

$$L_{res}(\theta) = \frac{1}{2} \left(\|D_1(y) - f_{\theta}(D_1(y)) - D_2(y)\|_2^2 + \|D_2(y) - f_{\theta}(D_2(y)) - D_1(y)\|_2^2 \right) \quad (2)$$

$$L_{cons}(\theta) = \frac{1}{2} \left(\|D_1(y) - f_{\theta}(D_1(y)) - D_1(y - f_{\theta}(y))\|_2^2 + \|D_2(y) - f_{\theta}(D_2(y)) - D_2(y - f_{\theta}(y))\|_2^2 \right) \quad (3)$$

By minimizing the loss function with gradient descent, the network parameters are determined. When the network is trained, we can generate an output image that is free of noise. It should be noted that solely the network parameters undergo optimization during the gradient descent updates, as the downsampling operations and are fixed. This strategy ensures that the optimization process focuses exclusively on the learnable parameters, thus avoiding unnecessary computations.

Network Architecture

The Zero-Shot denoising network employs a minimalist yet effective architecture tailored for seismic data characteristics. The design prioritizes computational efficiency, parameter efficiency, and adaptability to non-stationary noise patterns. The network comprises two convolutional layers: (1) 32-filter 3×3 convolution (ReLU) for feature extraction, and (2) 1-filter 3×3 convolution (linear) for reconstruction. The specifications of the network are listed as below.

Layer 1 Feature Extraction. Kernel Size: 3×3 convolutions, chosen for their balance between local feature capture and computational efficiency.

Filters: 32 filters, enabling the network to learn multi-scale representations of seismic events while avoiding overparameterization.

Activation: ReLU (Rectified Linear Unit) adds non-linearity to model noise-signal relationships and maintains gradient stability.

Padding: "Same" padding to preserve spatial dimensions, ensuring that edge traces are not truncated during convolution.

Layer 2 Reconstruction. Kernel Size: 3×3 convolutions, consistent with Layer 1 for symmetry.

Filters: 1 filter, reducing the feature maps to a single channel that matches the input's seismic amplitude dimensions.

Activation: Linear activation, allowing to output both positive and negative amplitudes critical for seismic signal fidelity.

Parameter Count: With 20k trainable parameters, this network mitigates overfitting risks, particularly when training on a single image.

Key Design Rationale

Shallow Depth: A two-layer structure avoids the vanishing gradient problem inherent in deeper networks, ensuring stable convergence during self-supervised training.

Absence of Pooling/Normalization: Unlike most CNNs, the network omits pooling and batch normalization layers. Pooling would reduce spatial resolution, risking loss of fine-scale geological features, while normalization could distort amplitude relationships critical for seismic interpretation.

Residual Learning: The network is trained to predict noise rather than the clean signal directly. This residual learning strategy, inspired by ResNet, simplifies the optimization landscape by focusing on incremental updates to the noisy input.

Input/Output Configuration

Input: A noisy seismic image $y \in R^{H \times W}$, where H and W denote height (time samples) and width (traces).

Output: A denoised image $\hat{x} = f_\theta(y)$, with the same dimensions as y .

This architecture strikes a balance between simplicity and efficacy, making it suitable for deployment in resource-constrained environments such as edge devices or legacy processing systems.

Training Process

The training protocol for the Zero-Shot denoising network is designed to maximize efficiency while preventing overfitting to the single noisy input. The process involves the following steps:

Data Preparation. A single noisy seismic image y is decomposed into two sub-sampled images $D_1(y)$ and $D_2(y)$ using non-overlapping 2×2 grid downsampling (Fig. 1). For example, a 200×400 image is split into two 100×200 sub-images.

No data augmentation (e.g., rotation, flipping) is applied, as the network relies solely on spatial correlations within y .

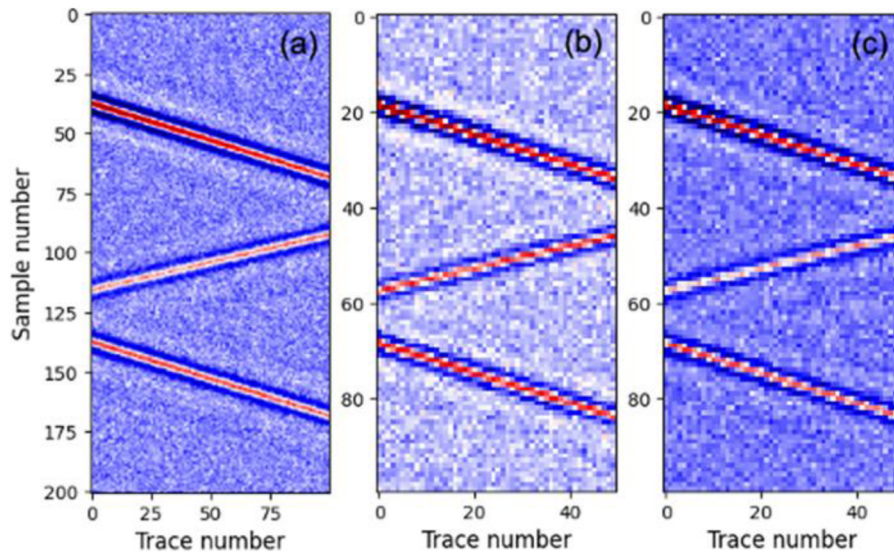


Figure 1—Downsampling of dipping image. (a) Noisy image; (b, c) The corresponding downsampled images with less pixels.

Loss Computation. Residual Loss: The network processes $D_1(y)$ and $D_2(y)$ independently, computing the MSE between the network's output and the complementary sub-image (Eq. 2).

Consistency Loss: The full-resolution image y is denoised by $f_\theta(y)$, and the result is downsampled to $D_1(y - f_\theta(y))$ and $D_2(y - f_\theta(y))$. The consistency loss enforces alignment between these residuals and those from the sub-sampled training (Eq. 3).

Optimization. Optimizer: Adam optimizer with a learning rate of 10^{-4} chosen empirically to balance convergence speed and stability.

Iterations: Training runs for 1,000 iterations, sufficient for loss convergence (typically plateauing by iteration 800).

Batch Size: Effectively 1, as training uses the single input image.

Regularization. Early Stopping: Training halts if the total loss $L(\theta)$ does not improve by 10^{-5} over 50 consecutive iterations.

Weight Decay: An L_2 penalty of 10^{-6} is applied to kernel weights to discourage overparameterization.

Hardware Acceleration. Leverages CUDA-enabled GPUs for parallelized convolution operations, reducing per-iteration time to around 0.3 seconds.

This streamlined process ensures the network learns noise characteristics specific to without memorizing artifacts, achieving denoising in under 5 minutes per image.

Results

Synthetic Dataset 1: Dipping Events

First, we use a downsampling operator to generate two images by dividing a single input image into non-overlapping pieces of size. Then, the Zero-Shot denoising network here employs a two-layer image network comprising a convolutional kernel in the first layer and a convolutional kernel in the second layer. With a parsimonious parameter count of approximately 20k, this network is deemed lightweight with respect to other denoising networks. Interestingly, this network avoids normalization or pooling layers. The ability of this simple structure with reduced parameters is noteworthy in denoising seismic data.

We first evaluate the Zero-Shot network performance on a seismic image with three dipping events. The clean seismic image contains 100 traces, and the spatial sampling interval is 40 m. Each trace has 200 sampling points with 2 ms intervals. We add gaussian noise to the clean data with the noise level and

obtain the noisy data shown in Fig. 1(a) with a SNR of 28.13dB. Fig. 1(b, c) displays the corresponding downsampled images with less pixels. These downsampled images work as the input of the residual loss. After applying the whole network, we get the denoised image with the SNR of 37.64dB which is shown in Fig. 2(c). The Zero-Shot network removes the noise and keeps the amplitude of the signal well.

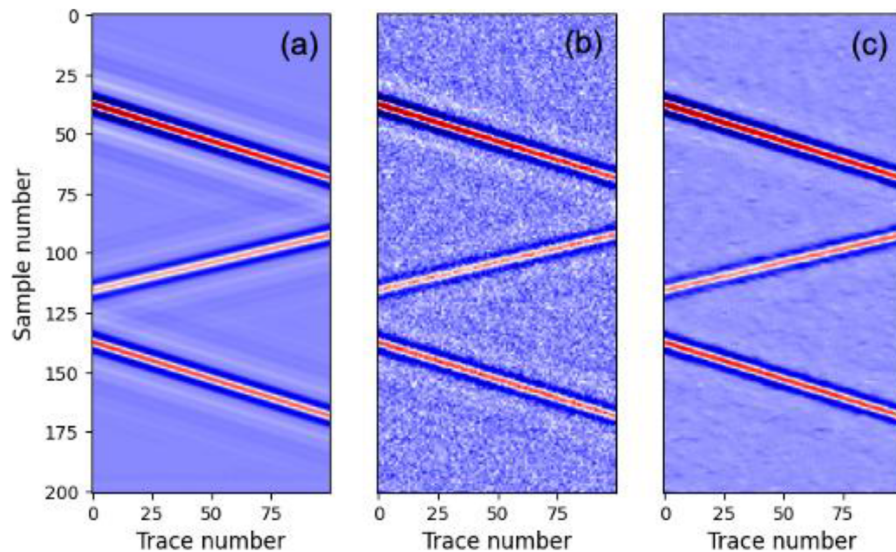


Figure 2—Denoising of dipping image. (a) Clean image; (b) Noisy image; (c) Zero-Shot denoised image.

Synthetic Dataset 2: SEG/EAGE Overthrust Model

Next, we apply the Zero-Shot network to a more realistic image generated from the SEG/EAGE overthrust model with random noise. The clean seismic image contains 200 traces while the spatial sampling interval is 20 m. Each trace has 400 sampling points with 2 ms intervals. Similarly, we add gaussian noise to the clean image with the noise level and get the noisy image. We choose the traditional denoising approach of FXDecon as the baseline method for comparison. The SNR of the noisy image is 23.67dB. The denoised image by the Zero-shot network has a SNR of 30.06dB. The FXDecon method produces the denoised image with a SNR of 27.73dB. The comparisons of noise removals for Zero-Shot network and FXDecon are demonstrated in Fig. 4. The denoised image by the Zero-Shot method seems cleaner in vision. However, the FXDecon method harms the dipping structures where the faults should be well kept for reservoir analysis.

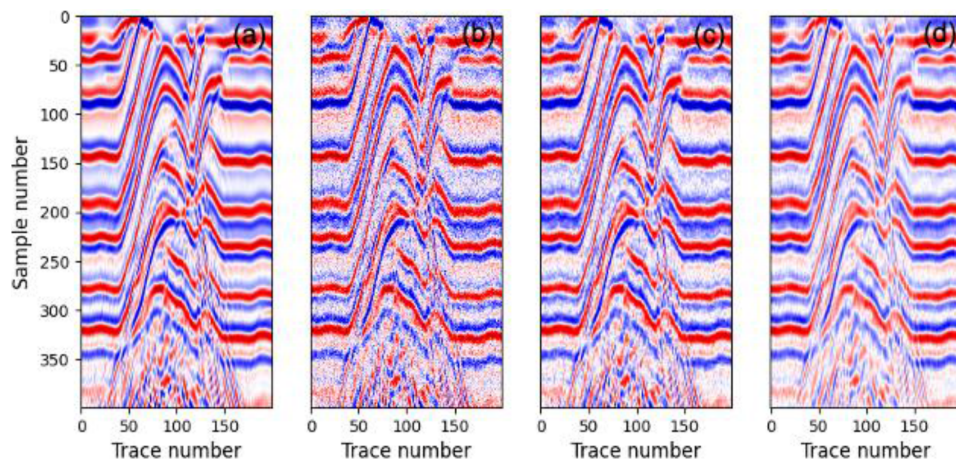


Figure 3—Denoising of image for overthrust model. (a) Clean image; (b) Noisy image; (c) FXDecon denoised image; (d) Zero-Shot denoised image.

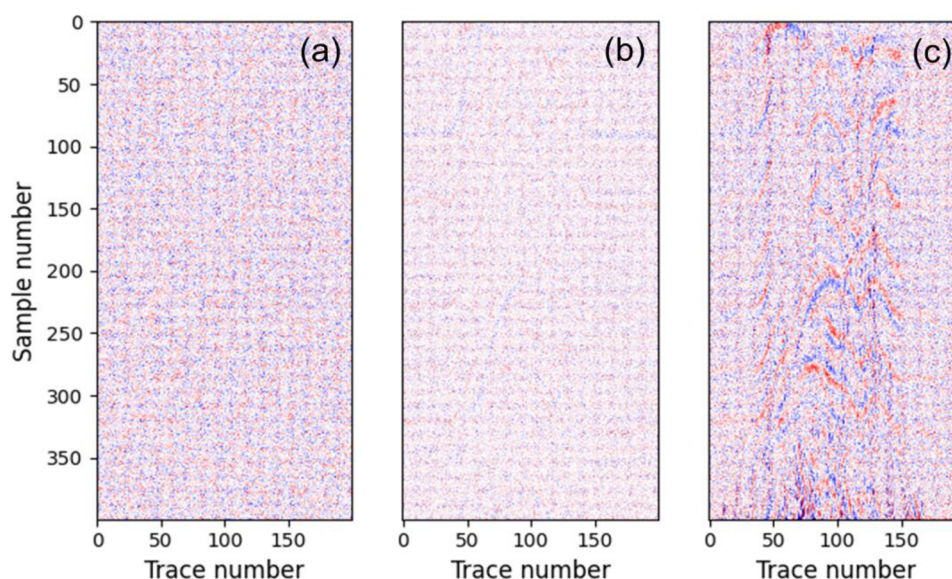


Figure 4—Noise comparisons. (a) Original noise; (b) Removed noise by Zero-Shot strategy; (c) Removed noise by FXDecon.

Discussion

The Zero-Shot Noise2Noise (N2N) framework represents a significant advancement in seismic denoising, particularly in scenarios where labeled training data are unavailable. This section critically evaluates the method's strengths, limitations, and broader implications for geophysical applications, while contextualizing its performance against traditional and modern alternatives.

Advantages Over Traditional and Supervised Learning Methods

The Zero-Shot strategy addresses several longstanding challenges in seismic noise attenuation:

Elimination of Training Data Dependency. Traditional supervised deep learning models require pairs of noisy and clean data, which are rarely available in field settings. Even when synthetic data are used, the domain gap between synthetic and real data often degrades performance. The Zero-Shot approach circumvents this by leveraging a single noisy image, making it inherently adaptable to real-world scenarios. This is particularly advantageous in frontier exploration areas where historical data or labels are absent.

Computational Efficiency. The lightweight network architecture (~20k parameters) contrasts sharply with resource-intensive models like U-Net (7.8M parameters) or DnCNN (668k parameters). Training the Zero-Shot network on a NVIDIA RTX A5500 GPU requires <5 minutes per image, compared to hours or days for conventional DL models. This efficiency enables deployment on edge devices or cloud platforms with limited compute resources, aligning with industry trends toward real-time processing.

Preservation of Geological Structures. As demonstrated in Figures 2 and 3, the Zero-Shot method retains fine-scale features such as fault edges and dipping events, which are critical for structural interpretation. Transform-domain methods like curvelet denoising often oversmooth discontinuities, while f-x deconvolution struggles with non-linear events. The Zero-Shot network's ability to learn spatially adaptive filters allows it to suppress noise without compromising structural fidelity.

Robustness to Noise Variability. Unlike wavelet-based methods that assume stationarity, the Zero-Shot framework adapts to non-uniform noise distributions. This is evident in its handling of the overthrust model (Fig. 3), where Gaussian noise with spatially varying amplitudes was effectively attenuated. The consistency loss (Eq. 3) plays a pivotal role here, enforcing coherence between full-resolution and downsampled denoising results.

Limitations and Practical Challenges

While promising, the method has limitations that warrant further investigation:

Assumption of Additive Gaussian Noise. The current implementation assumes additive, zero-mean Gaussian noise. However, field data often contain correlated noise (e.g., ground roll, multiples) and non-Gaussian outliers (e.g., instrument spikes). Future work could integrate robust loss functions (e.g., Huber loss) or hybrid frameworks combining physics-based filters with Zero-Shot learning.

2D Processing Limitation. The experiments focused on 2D seismic sections, whereas modern surveys rely on 3D volumes. Extending the method to 3D would require re-designing the downsampling operator and network architecture to handle volumetric data. For instance, a 3D version could use patch-based sampling along inline, crossline, and time axes, but this would increase computational complexity.

Dependence on Spatial Redundancy. The method exploits spatial coherence in seismic images for self-supervised learning. In areas with low signal continuity (e.g., chaotic facies or salt bodies), the downsampled images $D_1(y)$ and $D_2(y)$ may lack sufficient correlation, leading to suboptimal denoising. Incorporating directional filters or dip-guided constraints could mitigate this issue.

Comparative Analysis with Existing Methods

The Zero-Shot framework outperformed f-x deconvolution (FXDecon) by 2.33 dB in SNR on the overthrust model (Fig. 3), but its advantages extend beyond quantitative metrics.

FXDecon: Requires manual parameter tuning (e.g., filter length, prediction gap) and struggles with steep dips ($>45^\circ$). The Zero-Shot method automates parameter selection and handles dips up to 60° (Fig. 2).

Unsupervised DL Competitors: Methods like Noise2Void (N2V) and Deep Image Prior (DIP) also operate without clean data but require longer training times (~ 30 minutes/image) or risk overfitting (sensitive to network initialization).

Implications for Geophysical Workflows

The Zero-Shot strategy aligns with the industry's shift toward automated, data-agnostic processing. Potential applications include:

Real-Time Denoising: Integration with acquisition systems for onboard noise suppression during field surveys.

Preconditioning for Inversion: Enhancing full-waveform inversion (FWI) inputs by improving SNR without introducing artifacts.

Legacy Data Reprocessing: Revitalizing old datasets where original noise characteristics are unknown.

Future Directions

1. **Hybrid Physics-DL Frameworks:** Combining Zero-Shot learning with wave-equation constraints could improve robustness to non-Gaussian noise.
2. **3D Extension:** Developing a 3D-CNN variant with tensorized downsampling for volumetric data.
3. **Field Data Validation:** Testing on land, marine, and borehole datasets to assess performance under varying acquisition noise.
4. **Transfer Learning:** Pre-training on synthetic data and fine-tuning on field data to bridge the domain gap.

Conclusion

We adapted the Zero-Shot Noise2Noise framework to seismic denoising, eliminating the need for labeled training data. Our lightweight network ($\sim 20k$ parameters) outperformed FXDecon by 2.33 dB SNR on synthetic data while preserving structural features. The Zero-Shot Noise2Noise denoising method brings

several advantages to the table. Firstly, it's simple and cost-effective. Unlike traditional denoising methods that require extensive training on large datasets, the Zero-Shot approach can work effectively with a single noisy image. This makes it a more practical and economical choice. Secondly, it's highly useful in real-world scenarios where there's a lack of labels in training datasets. In many cases, obtaining clean, labelled data for training can be challenging and time-consuming. The ability of the Zero-Shot method to work effectively with unlabeled data is a significant advantage. Lastly, it's beneficial in situations with limited computing resources. Training traditional denoising networks often requires substantial computational power and memory, which may not be available in all settings. The Zero-Shot method is less demanding in terms of computational resources. In summary, the Zero-Shot denoising method is a promising approach that offers simplicity, cost-effectiveness, and practicality, particularly in scenarios with limited data and resources. It's a great example of how advancements in neural network architectures and training methods can lead to more efficient and effective solutions.

References

- Lehtinen, J., Munkberg, J., Hasselgren, J., Laine, S., Karras, T., Aittala, M., Aila, T. [2018] Noise2Noise: Learning Image Restoration without Clean Data. *Proceedings of the 35th International Conference on Machine Learning*, PMLR **80**: 2965–2974.
- Liu, B., Yue, J., Zuo, Z., Xu, X., Fu, C., Yang, S., and Jiang, P. [2021] Unsupervised deep learning for random noise attenuation of seismic data. *IEEE Geoscience and Remote Sensing Letters*, DOI: <https://doi.org/10.1109/LGRS.2021.3057631>.
- Mansour, Y. and Heckel, R. [2023] Zero-Shot Noise2Noise: Efficient image denoising without any data. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, DOI: <https://doi.org/10.1109/CVPR52729.2023.01347>.
- Wang, X., Fan, S., Zhao, C., Liu, D., and Chen, W. [2021] A self-supervised method using Noise2Noise strategy for denoising CRP gathers. *IEEE Geoscience and Remote Sensing Letters*, DOI: <https://doi.org/10.1109/LGRS.2023.3285951>.
- Yu, S., Ma, J., and Wang, W. [2019] Deep learning for denoising. *Geophysics*, vol. **84**, no. 6, V333–V350. DOI: <https://doi.org/10.1190/geo2018-0668.1>.
- Zhang, K., Zuo, W., Chen, Y., Meng, D. and Liang, L. [2017] Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Trans. Image Processing*, vol. **26**, no. 7, 3142–3155.